T97700 - Large-Prime Parity Boundary - repository archival PDF

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lheadLarge-prime parity boundary
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titlebfseries The Large-Prime Parity Boundary2mm]
large A No-Go for LAPBR67 and an Exact Signed Type-II Bellman Frontier
Agentic Polymath Project
18 August 2026
begincenter
fcolorboxred!70!blackwhiteparbox0.91textwidthsmall
Status. The adaptive small-prime cube is retained. The proposed
nonnegative large-prime residual is disproved. The paper replaces it with one
exact largest-prime Bellman identity and one open signed Type-II estimate.
The Riemann Hypothesis is not proved.
endcenter

ABSTRACT
A recent factor--67 parity proposal proves positivity of the source histories
of length below an adaptive even depth and supported on primes
p <= (log X)^1/4. It asks that the complementary histories be
nonnegative. We show that this residual theorem, called LAPBR67, is
false. A product-safe elementary symmetric-sum argument applies at the
published depth L=O(logloglog X): the final layer may be restricted to
primes at most X^1/(2L), while all preceding layers admit a full reciprocal-
prime upper bound. Mertens' theorem then makes the last odd layer dominate, so
the complete depth current is
negative, and the residual is more negative than the small-prime cube is
positive.

The repair is to resum every large-prime depth. We prove positivity of the
complete small-prime cube, assign every remaining history to its largest prime,
and obtain an exact Bellman identity. The terminal range p>X/Z is
o(U_Z(X)). The only remaining quantity is a one-sided signed bilinear
correlation between the largest prime and a M"obius tail. We formulate the
minimal estimate BLPTE67; it implies the annular scalar inequality and
therefore the Riemann Hypothesis through the exact reciprocal-zeta Mellin
consumer. BLPTE67 remains open.

■Genealogy and notation■==
Let b(Y) be the complete P_61 annular 5:3 scalar from the repaired
finite theorem. It is nonnegative and, once the fixed P_61 activation
range has been passed,
labeleq:base-asymp
b(Y)=a_*sqrt Y+c_*+O(Y^-3/2),

a_*=12PROD_p <= 61(1-frac1p)>0.

For squarefree rough histories define
labeleq:Br
B_r(X)=SUM_substackP^-(m) >= 67omega(m)=r
frac1sqrt mb(X/m),

with zero extension below support. The exact depth-L current is
labeleq:CL
C_L(X)=SUM_r=0^L-1(-1)^rB_r(X).

■Product-safe domination of the last layer■
==
Put

Lambda_X=SUM_67 <= p <= Xfrac1p=loglog X+O(1).

The no-go needs no uniform Sathe--Selberg theorem. It follows from elementary
symmetric sums after restricting only the final layer to a product-safe prime
range.

LEMMA[Product-safe layer comparison]labellem:layers
Let r=O(logloglog X), put Y=X^1/(2r), and set

A_Y=SUM_67 <= p <= Yfrac1p.

Then

B_r(X) >= c_0sqrt XfracA_Y^rr!(1-o(1)),

whereas

SUM_j<rB_j(X)
<= 2C_0sqrt XfracLambda_X^r-1(r-1)!.

If additionally rlog r/Lambda_X -> 0, then

fracB_r(X)SUM_j<rB_j(X)longrightarrowinfinity.

PROOF.
Equation~eqrefeq:base-asymp and the finite initial range give constants
c_0,C_0,Y_0>0 with

b(u) >= c_0sqrt u (u >= Y_0),
0 <= b(u) <= C_0sqrt u (u >= 1).

Every product of r primes at most Y is at most sqrt X, so the first
inequality may be used on the entire selected final layer. Let e_r(Y) be the
elementary symmetric sum of the weights 1/p. Expanding A_Y^r over ordered
tuples and applying a union bound to the event that some pair of coordinates
coincides gives

r!e_r(Y)
>= A_Y^r-binomr2
(SUM_p >= 67frac1p^2)A_Y^r-2.

This proves the lower bound for B_r.

For earlier layers, Maclaurin's inequality gives
B_j(X) <= C_0sqrt XLambda_X^j/j!.
Since r=o(Lambda_X), the displayed terms increase with j<r, so their
sum is at most twice its last term.

Mertens' theorem gives

A_Y=Lambda_X-log(2r)+O(1).

It follows that

fracB_r(X)SUM_j<rB_j(X)
ggfracA_Yrleft(fracA_YLambda_Xright)^r-1.

Now r=o(Lambda_X) and
rlog r/Lambda_X -> 0, so the right side tends to infinity.
QED.

■The published residual is negative■==
THEOREM[LAPBR67 no-go]labelthm:no-go
At the adaptive depth of PR~#578,

C_L(X)<0

for all sufficiently large X. If S_X>0 is its proved small-prime cube and
mathcal L_X is the claimed residual, then

mathcal L_X<-S_X<0,

and in fact -mathcal L_X/S_X -> infinity.

PROOF.
The chosen depth is O(logloglog X), so it satisfies the hypotheses of
Lemma~reflem:layers. Since L is even, r=L-1 is odd. The final odd layer
dominates the sum of all preceding layers, proving C_L<0.
The identity C_L=S_X+mathcal L_X gives the first residual claim.
The predecessor gives S_Xllsqrt X, whereas the proof of Lemma~reflem:layers
gives B_r/sqrt X -> infinity, proving the ratio claim.
QED.

■The complete small-prime cube■==
Put Z=(log X)^1/4 and Q_Z=PROD_67 <= p <= Zp. Define

mathcal B_Z(X)=SUM_d | Q_Zfracmu(d)sqrt db(X/d),
qquad U_Z(X)=fracmathcal B_Z(X)sqrt X.

THEOREM[Complete small-prime cube positivity]labelthm:small-cube
For sufficiently large X,

mathcal B_Z(X)=
a_*sqrt XPROD_67 <= p <= Z(1-frac1p)
+c_*PROD_67 <= p <= Z(1-frac1sqrt p)+o(1),

so mathcal B_Z(X)>0 and U_Z(X)asymp1/log Z.

PROOF.
The largest divisor of Q_Z is exp((1+o(1))Z)=X^o(1),
so the base asymptotic is uniform over all d | Q_Z. Substitution and exact
divisor factorization give the two products. The total error is

O(X^-3/2PROD_67 <= p <= Z(1+p))=o(1).

The square-root term is positive and of order sqrt X/log Z. The constant
term is exponentially smaller.
QED.

■Largest-prime Bellman ownership■==
Adjoin active primes Z<p_1<cdots<p_k <= X/2 in increasing order. The exact
normalized finite-prime recurrence is

U_i(Y)=U_i-1(Y)-p_i^-1U_i-1(Y/p_i),   U_0=U_Z.

THEOREM[Largest-prime Bellman identity]labelthm:bellman
The complete normalized scalar is
labeleq:bellman
U_rm full(X)=U_Z(X)-
SUM_Z<p <= X/2frac1pU_<p(X/p).

Moreover
labeleq:child-expansion
U_<p(Y)=
SUM_substackv squarefreeZ<P^-(v), P^+(v)<p
fracmu(v)vU_Z(Y/v).

PROOF.
Telescope the recurrence. Expanding the previously adjoined primes in each
child yields~eqrefeq:child-expansion. Every nonempty subset occurs exactly
once, in the summand indexed by its largest prime.
QED.

The predecessor's truncated residual also has the exact Duhamel identity

P_<L(R)b-P_<L(S)b
=-SUM_a+b <= L-2(-1)^a+bS^a(R-S)R^b b,

which displays its short remaining histories. Theorem~refthm:no-go shows
why they must be resummed rather than signed separately.

■Type I and Type II■==
Split the Bellman budget in~eqrefeq:bellman at p=X/Z:

mathfrak I_Z(X)=SUM_X/Z<p <= X/2frac1pU_Z(X/p), mathfrak T_Z(X)=SUM_Z<p <= X/Zfrac1pU_<p(X/p).

The Type-I equality is exact because X/p<Z, so no prime above Z is active.

PROPOSITION[Terminal Type-I closure]labelprop:typei

|mathfrak I_Z(X)|llfrac(log Z)^2log X=o(U_Z(X)).

PROOF.
For 2 <= y <= Z, unsigned divisor mass gives
|U_Z(y)|lllog(2y). The reciprocal-prime mass of
(X/Z,X/2] is O(log Z/log X) by Mertens' theorem.
QED.

The remaining term is the exact signed Type-II correlation
labeleq:typeii
mathfrak T_Z(X)=
SUM_Z<p <= X/Zfrac1p
SUM_substackv squarefreeZ<P^-(v), P^+(v)<p
fracmu(v)vU_Z(X/(pv)).

No absolute values have been taken in~eqrefeq:typeii.

DEFINITION[BLPTE67]
The balanced large-prime Type-II estimate is

mathfrak T_Z(X) <= U_Z(X)-mathfrak I_Z(X)

at every sufficiently large admissible state.

■Conditional completion■===
THEOREM[Exact remaining theorem]labelthm:conditional

BLPTE67=>RH.

PROOF.
The Bellman identity and BLPTE67 give nonnegativity at each state.
Child endpoints are smaller, so finite induction starts from the directly
checked base range. At the root one obtains eventual nonnegativity of

mathcal A_X=5[c_X(2)-c_X/4(2)]+3[c_X(3)-c_X/4(3)].

Its Mellin transform is

(1-4^-s)[frac6s^2-
frac3(1-2^-z)(2-2^-z)s^2zeta(z)],
z=s+tfrac12.

The elementary factors do not cancel a zeta zero in Re z>0. Landau's
theorem excludes zeros to the right of the critical line, and the functional
equation supplies the reflection.
QED.

■Proof boundary■===
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Statement & Status
midrule
Adaptive small-prime cube & retained
LAPBR67 residual nonnegativity & false
Product-safe last-layer domination & proved from symmetric sums and Mertens
Complete small-prime cube & proved positive
Largest-prime Bellman identity & exact
Terminal Type I & closed
Signed balanced Type II, BLPTE67 & open / RH-bearing
Riemann Hypothesis & unproved
bottomrule
endlongtable

■Conclusion■===
The large prime does simplify the source, but not by making a shallow residual
positive. It supplies a unique owner, a strict endpoint reduction, a terminal
Type-I range, and a canonical prime-versus-M"obius Type-II form. The original

residual sign is false. The exact remaining theorem is the one-sided estimate BLPTE67; proving it, without discarding the M"obius signs, would finish this lane.